ONE-PAGE BRIEF

Post-quantum cryptography review: agave @ 7700cb3128c1

Client: Anza (Solana) — public repository (passive review) · Source: github.com/anza-xyz/agave, commit `7700cb3128c1` · Reviewed: 2026-11-04 18:21 UTC · Risk profile: Crypto exchange, custodian or wallet

POST-QUANTUM MATURITY

Level 0 of 4

no post-quantum cryptography

FINDINGS

0 Critical **15** High **32** Medium **34** Low **27** Info

QUANTUM TIMELINE (MOSCA: X + Y − Z)

Key establishment: 10 + 5 − 8 = 7 → **too late**

Signatures: 10 + 5 − 8 = 7 → **too late**

Next step: Inventory where classical public-key cryptography protects long-lived data and keys, and choose the first migration: usually a hybrid TLS key exchange and wrapped keys.

WHAT MATTERS MOST

High QSI-2026-11545	**openssl 0.10.64 is affected by GHSA-8c75-8mhr-p7r9 (CVE-2026-41678)** What to do: Upgrade to a fixed version (0.10.78).
High QSI-2026-11546	**openssl 0.10.64 is affected by GHSA-ghm9-cr32-g9qj (CVE-2026-41681)** What to do: Upgrade to a fixed version (0.10.78).
High QSI-2026-11547	**openssl 0.10.64 is affected by GHSA-hppc-g8h3-xhp3 (CVE-2026-41898)** What to do: Upgrade to a fixed version (0.10.78).
High QSI-2026-11549	**openssl 0.10.64 is affected by GHSA-pqf5-4pqq-29f5 (CVE-2026-41676)** What to do: Upgrade to a fixed version (0.10.78).

77 more in the full report.

WHAT THE QUANTUM-VULNERABLE CRYPTOGRAPHY PROTECTS

Role	Algorithms	What to do
Medium channel key exchange (TLS, Noise, handshakes)	TLS ECDHE	Move the handshake to a hybrid key exchange (X25519MLKEM768 in TLS 1.3; a hybrid KEM in Noise-based protocols)
Medium threshold signing and MPC protocols	Pedersen commitment	Track post-quantum threshold signatures (NIST IR 8214C) and keep the protocol layer ready to swap the...
Medium transaction and message signing	Ed25519, ECDSA, BN254 (alt_bn128)	Follow each chain's post-quantum roadmap, avoid public-key reuse, and keep signers ready to add post-quantum...
Medium role not determined	Ed25519, BN254 (alt_bn128), Bulletproofs	Classify each usage, record data lifetimes, and plan migration to ML-DSA (FIPS 204) or SLH-DSA (FIPS 205)...

KNOWN VULNERABLE COMPONENTS

Advisories for the versions in use: 78; unlikely to apply to this build: 16 (a Cargo feature that is off, affected functions not named in the code, or a platform that is not built); to review: 62.

Static review of source code and manifests; files analysed: 1293. No requests were sent to the owner's systems.

Full report, content SHA-256: `0bd9b0a0f4a11ff2fadcc1131f86ad7cdd40e0964340c4b37ac351a57d2386d5`

Full report signature (report.sig.json): Ed25519 key, SHA-256 fingerprint `5b3cf9c9bd0fa54447e2b5dfa7a6bab569bd0c666fa46949599e5c348d17083e`. bundle.sig.json lists the SHA-256 of every file in this bundle and is signed too; verify-bundle checks it offline.

Preliminary findings need confirmation in the code; the full report gives the evidence, the limits of each check and how to re-test.